

# IKT215 Exercise 10:

## Principal Component Analysis

April 9<sup>th</sup>, 2025

*Exercise sessions, while not mandatory, are an integral part of your learning experience and exam preparation. Some exam questions will be based on tasks covered during these sessions. Although you have the flexibility to complete exercises at home and ask questions later (preferably within one week), I strongly recommend attending the sessions in person for immediate support and guidance. This allows you to benefit from direct interaction and real-time problem-solving. I respond promptly to emails ([manuel.s.mathew@uia.no](mailto:manuel.s.mathew@uia.no)).*

### Task 1: Calculations

Below is a small dataset with 5 types of fruit, each described by two features:

- Calories per 100g: The number of calories in 100 grams of fruit.
- Sugar per 100g: The amount of sugar (in grams) in 100 grams of fruit.

| Fruit      | Calories per 100g | Sugar per 100g |
|------------|-------------------|----------------|
| Apple      | 52                | 10             |
| Banana     | 89                | 12             |
| Orange     | 47                | 9              |
| Grapes     | 69                | 16             |
| Strawberry | 32                | 5              |

1. Scale the dataset, if necessary.
2. Calculate the PCA for this small dataset:
  - a. Center the data,
  - b. Compute the covariance matrix,
  - c. Find the eigenvalues and eigenvectors,
  - d. Calculate the first and second principal components.
  - e. Calculate the percentage information in each of the principal components.
  - f. Using matplotlib plot the original data (scatter plot) and the eigenvectors.

### Task 2: PCA on Iris Dataset

#### 1. PCA Implementation & Variance Analysis

- a. Perform PCA using standard Python libraries. Calculate and report the percentage of variance explained by each principal component.
- b. Determine the minimum number of principal components required to retain **95%** of the original variance.

- c. Plot the cumulative explained variance ratio curve.

## 2. Visualization

Plot a labeled scatter plot of PC1 vs PC2, colored by iris species. Include axis labels with explained variance percentages.

## 3. Reconstruction Error Analysis

- a. Reconstruct the dataset using only the first two principal components.
- b. Calculate and report the mean squared error (MSE) between the original and reconstructed data.

## 4. Preprocessing Impact Test

- a. Repeat PCA **without standardization** (use raw data).
- b. Compare the explained variance ratios between standardized and non-standardized PCA.
- c. Explain why standardization is critical for PCA in this dataset